Assignment 10 - Test and Validation Plan Draft

Test and Validation Plan (version 1.0, November 4, 2026)

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Source: https://github.com/AllyOI/model-recommend/blob/main/Test_Plan.md · Senior Design 1, Fall 2026

Test and Validation Plan — Model Recommend

Project: Model Recommend (Concentrate AI) **Team:** Allison Oh, Aditya Pawar **Test lead (document owner):** Allison Oh. Each component owner wrote the cases for their component: Aditya Pawar for the Recommendation Server, LLM adapter and Authentication; Allison Oh for the Recommendation Client, Report Store, Notifications and Configuration. **Version:** 1.0 (2026-11-04). Traces to *Requirements_Document.md* v1.0 and *Technical_Specification.md* v1.1, whose traceability matrix names the test-case groups TC-01 to TC-29 used here.

Numbering. Each Week 9 matrix ID (for example TC-06) is a group; the cases in it are TC-06.1, TC-06.2 and so on. No group was added or renumbered. The sub-numbering is logged in the Technical Specification's version log (v1.1). Every case below has status **Not Run**: implementation starts in CS 5002 (milestone M0, 2027-01-20).

1. Scope

This cycle is the CS 5002 build, January to April 2027 (milestones M0 to M7 in *Planning/Plan.md*). Requirements are tested in risk order: anything that sends client data out or writes stored data first (REQ-03, REQ-04, REQ-05, NFR-08, NFR-09), then the core run (REQ-06, NFR-14, REQ-01, REQ-02, NFR-05), then everything else.

In scope.

Component (D0/D2)	Requirements tested	Owner
Recommendation Server: validation, run lifecycle, A1 dispatch, A2 ranking, review, selection	REQ-01, REQ-02, REQ-03, REQ-04, REQ-06, REQ-08, REQ-09, REQ-13; NFR-05, NFR-07, NFR-14, NFR-15	Aditya Pawar
LLM adapter (`callVendor`) and usage log	REQ-06, REQ-11; NFR-14	Aditya Pawar
Authentication glue	NFR-08	Aditya Pawar
Report Store (schema, repository methods, indexes)	REQ-05, REQ-07, REQ-12; NFR-06, NFR-15	Allison Oh
Recommendation Client (screens S1–S6)	REQ-03 (client checks), REQ-05, REQ-07; NFR-10, NFR-11, NFR-13	Allison Oh
Notifications glue	REQ-09, REQ-10	Allison Oh
Configuration, logging and CI	NFR-07, NFR-09 (log redaction), NFR-10	Allison Oh
Constraints	CON-01 to CON-04 through the cases listed in Section 4; CON-05 at milestone reviews	Both

Out of scope this cycle, and why.

Item	Reason
Could requirements REQ-14, REQ-15, REQ-16, NFR-12, NFR-16	Deferred features. Their cases (TC-14, TC-15, TC-16, TC-24, TC-28) run only if the feature ships by M5; otherwise they stay Not Run and are reported as deferred, not failed.
Won't items W-01 to W-09 (Requirements Document, Section 7)	Not built, so not tested: self-hosted models, public comparisons, client-set weights, learned ranker, tool-calling and agentic benchmarks, a broker, admin screens, mobile app, multi-region.
Known limitation: one test location	Every latency number is measured from one region (CON-05, Tech Spec Section 5).
Known limitation: rule-based sanitizer	TC-21 tests the 200-item corpus NFR-09 defines; formats outside it are a known gap until OQ-05 is decided.
Known limitation: fewer than 3 models	TC-01.4 checks the documented behavior; whether AC-01.1 changes is OQ-01.
Third-party components not retested	OpenAI and Anthropic APIs, the OIDC identity provider, PostgreSQL itself, the hosting platform, the email provider, Slack, and libraries (Fastify, zod, p-limit, p-retry, jose, Kysely, pino). We test our use of them through mocks and staging, not the components themselves.

2. Verification and Validation Strategy

Verification (built right, against the spec) is done by the team for every functional requirement and every non-functional requirement with a measured threshold: REQ-01 to REQ-16 and NFR-05 to NFR-10, NFR-13 to NFR-16. Almost all of it is automated and runs in CI against the mock vendor server, so results do not depend on vendor uptime or cost (CON-01). A smaller set repeats on staging with real vendor APIs (TC-06.4, TC-17.2, TC-26 subset).

Validation (the right product, for the stakeholders) is done with people outside the team:

- **Product managers** (US-01, US-02): usability sessions for NFR-11 with 5 first-time users, run first on staging and again in the pilot with Concentrate AI product managers.
- **Solutions engineer and advisor** (US-01, CON-03): ranking validation TC-01.5, where a Concentrate AI solutions engineer ranks the same candidates by hand and we compare.
- **IT security reviewer** (US-04): NFR-12 session, if built.
- **Todd Liebermann** (CON-04): sign-off on reasoning wording (TC-29) and the M4 demo.
- **Jillian Aurisano**: reviews this plan and the M6 test report.

Model-based parts. The quality grade comes from an LLM grader call, so ranking has run-to-run noise. Variables: brief, approved model set, prompts per model (5). Metrics: top-ranked model and fit score. Trials: 5 identical runs per brief on staging. Tolerance: the same top-1 model in at least 4 of 5 trials and a fit-score standard deviation of at most 5 points per model (TC-01.6).

Test pyramid.

Layer	Approx. count	Tool	Coverage focus
Unit	120	vitest	Brief validation, A2 scoring and tie-breaks, reasoning template, sanitizer, limiter state machine, config validation, cost arithmetic
Integration	45	vitest + Docker PostgreSQL 16 + mock vendor	Endpoints E1–E12 with the database, auth matrix, run lifecycle, notifications outbox, usage log
System (end to end)	25	Playwright through the Client	UC-01 and UC-02 flows, error messages, review and selection
Load and fault	6 scenarios	k6, Toxiproxy, mock vendor modes	NFR-05, NFR-06, NFR-07, NFR-14, NFR-15
Acceptance and validation	3 sessions + M4 demo	Scripted sessions, observer sheet	NFR-11, NFR-12, TC-01.5, TC-29

3. Success Criteria (Metrics)

Metric	Target threshold	How it is measured	Req. ID
Run completion time	≥ 95% of 20 consecutive runs at realistic size reach a terminal status ≤ 90 s after 202	`runs.finished_at − runs.created_at` for each run	NFR-05
Fit-score correctness	100% of 30 fixture models match the reference spreadsheet exactly (integer)	Unit test against `fixtures/a2_reference.csv`	REQ-01
Ranking validation (validation)	System top-1 is in the solutions engineer's top 3 for ≥ 4 of 5 briefs	Blind comparison, TC-01.5	REQ-01, CON-03
Ranking stability (validation)	Same top-1 in ≥ 4 of 5 trials; fit-score SD ≤ 5 points	5 staging runs per brief, TC-01.6	REQ-01
Reasoning accuracy	100% of numbers in reasoning text equal the stored entry values	String parse vs `report_entries`	REQ-02
Rate-limit reaction	`delayed` ≤ 5 s after the 2nd 429; 0 calls during max(60 s, Retry-After)	Server log timestamps vs mock request log	REQ-06
Approval activation	List applied to every run started ≥ 60 s after save	Mock request log per vendor	REQ-05
Unapproved-vendor calls	0	Mock request log	REQ-04
Read-endpoint latency	p95 ≤ 500 ms at 20 virtual users for 5 min, 10,000 reports	k6 report + `Server-Timing` header	NFR-06
Calls per run and quotas	≤ 2 × models × prompts + 1 retry per failure; 4th active run → 409; 21st run/hour → 429	Count rows in `vendor_calls`; API responses	NFR-07
Usage-log completeness and cost accuracy	100% of calls logged ≤ 2 s after the call; cost = tokens × price within $0.000001	Compare mock log with `vendor_calls`	REQ-11
Cancellation	0 vendor calls start > 5 s after cancel; no report stored	Mock log, `reports` table	REQ-13
Auth matrix	100% of role × endpoint × token cases return the expected code	Integration test table	NFR-08
PII removal	100% of 200 seeded items absent from outbound prompts; 0 prompts, completions or tokens in logs	Regex scan of mock log and log files	NFR-09
Transport and secrets	TLS ≥ 1.2 only; 0 secrets found by gitleaks; token absent from browser storage	testssl.sh, CI scan, Playwright storage check	NFR-10

Metric	Target threshold	How it is measured	Req. ID
Product-manager task success	≥ 4 of 5 users finish within 3 min, unaided	Observer stopwatch and sheet	NFR-11
Reviewer task success	≥ 4 of 5 users finish within 2 min, unaided	Observer stopwatch and sheet	NFR-12
Error-message clarity	0 raw error codes shown for the 17 4xx codes	Playwright screenshot check	NFR-13
Vendor isolation	Other vendors scored in 20 of 20 runs for each of 4 failure modes	Mock modes, run results	NFR-14
No silent loss	Interrupted runs `failed` ≤ 30 s after restart; 0 non-terminal runs	SQL query after restart	NFR-15
Notification retry	5 retries at 30/60/120/240/480 s; 0 duplicates per key	`notifications` table	NFR-16

4. Test Cases

All cases use the seeded data in Section 5 (client companies Acme, Beta, Gamma; users `pm_acme`, `rev_acme`, `pm_beta`, `pm_gamma`, `se_1`, `se_2`, `op_1`; vendors V1–V4). "Mock" is the mock vendor server; its request log records every call it receives. "Standard brief" is `{budget: 3.00, latency_ms: 2000, task_description: "Summarize customer support tickets into two sentences."}`. Status for every case: **Not Run**.

TC-01: Ranked shortlist (REQ-01, Must; owner Aditya Pawar)

ID	Description	Steps	Expected Result	Status
TC-01.1	Happy path ranking (REQ-01)	1. Seed; set mock to ok with grades file `grades_a.json`. 2. As `pm_acme`, POST E1 with the standard brief and a new Idempotency-Key. 3. Poll E5 every 2 s until terminal.	Status `released`; 10 entries; every ok entry has an integer `fit_score` 0–100; ranks 1..10 unique; order equals `expected_order_a.json`.	Not Run
TC-01.2	Tie-break order (REQ-01, edge)	1. Load fixture with models M3 and M7 given identical samples, M3 cheaper; M8 and M9 identical in samples and price. 2. Run A2 unit test.	M3 ranks above M7; M8 above M9 (name order); repeated 10 times, same order.	Not Run
TC-01.3	All models over budget (REQ-01, edge)	1. As `pm_acme`, POST E1 with budget 0.01, otherwise standard. 2. Poll E5.	All entries `within_limits: false`, still ranked 1..10; each reasoning names the ceiling missed.	Not Run
TC-01.4	Fewer than 3 rankable models (REQ-01, failure)	1. Set mock V1, V2, V3 to 401. 2. Run standard brief as `pm_acme`.	Only V4's models ranked (2); warning `fewer_than_3_models`; no unscored entry has a rank (OQ-01 behavior).	Not Run
TC-01.5	Ranking validation (US-01, CON-03, validation)	1. Pick 5 pilot briefs. 2. Solutions engineer ranks the candidate models by hand without seeing results. 3. Compare with the system's top-1.	System top-1 is in the engineer's top 3 for ≥ 4 of 5 briefs.	Not Run
TC-01.6	Ranking stability (REQ-01, validation)	1. On staging, run the same brief 5 times. 2. Record top-1 and fit scores.	Same top-1 in ≥ 4 of 5 runs; SD of each model's fit score ≤ 5.	Not Run

TC-02: Reasoning text (REQ-02, Must; owner Aditya Pawar)

ID	Description	Steps	Expected Result	Status
TC-02.1	Reasoning numbers (REQ-02)	1. Run A2 on fixture `a2_reference.csv` with budget 3.00, latency 2000. 2. Parse each reasoning string.	Each contains quality (2 decimals), "$x.xx per 1,000 requests (y% of your $3.00 ceiling)" and "median latency L ms (z% of your 2,000 ms ceiling)", all equal to stored values.	Not Run
TC-02.2	Same input, same text (REQ-02)	1. Run A2 twice on identical samples. 2. Store a report, reopen it via E2 after 1 day (clock advanced).	Strings byte-identical across both runs and the reopened report.	Not Run
TC-02.3	Unscored entry (REQ-02, edge)	1. Set mock V2 to 429 always. 2. Run standard brief.	V2 entries have `fit_score: null` and reasoning stating the vendor was rate-limited and paused.	Not Run

TC-03: Brief validation (REQ-03, Must; owner Aditya Pawar)

ID	Description	Steps	Expected Result	Status
TC-03.1	Missing field (REQ-03, AC-01.2)	1. Note `runs` count and mock log size. 2. POST E1 as `pm_acme` without `budget`. 3. Repeat without `latency_ms`, then without `task_description`.	Each returns 400 `invalid_brief` with `field` set to the missing name; `runs` count and mock log unchanged.	Not Run
TC-03.2	Boundaries (REQ-03, edge)	POST E1 with budget 0.01, 10000, 0.009, 10000.01; latency 100, 60000, 99, 60001; task 10, 2000, 9, 2001 characters (12 requests, other fields standard).	The six in-range values return 202; the six out-of-range values return 400 naming the field; 0 vendor calls for rejected ones.	Not Run
TC-03.3	Bad input in the Client (REQ-03, NFR-13)	1. In S1 type budget "3,00", latency "1500.5", a 2,001-character task. 2. Try Submit. 3. Send the same values directly to E1.	S1 shows the field messages and Submit stays disabled; direct API calls return 400.	Not Run

TC-04: Approved vendors only (REQ-04, Must; owner Aditya Pawar)

ID	Description	Steps	Expected Result	Status
TC-04.1	Only approved vendors called (REQ-04)	1. Set Acme's list to V1, V2. 2. Run standard brief. 3. Filter mock log by vendor.	Benchmark and grader calls go only to V1 and V2; 0 calls to V3, V4.	Not Run
TC-04.2	No list (REQ-04, failure)	As `pm_beta` (no approvals) POST E1.	409 `no_approved_vendors`; 0 mock calls; S1 shows the "ask your IT reviewer" message.	Not Run
TC-04.3	Expired list (REQ-04, edge)	1. Set Acme's `expires_at` to 1 minute ago. 2. POST E1.	409 `no_approved_vendors`; 0 mock calls.	Not Run
TC-04.4	Store down, fail closed (REQ-04, failure)	1. Cut the database connection with Toxiproxy. 2. POST E1.	503 `store_unavailable`; 0 mock calls.	Not Run

TC-05: Vendor approval (REQ-05, Must; owner Allison Oh)

ID	Description	Steps	Expected Result	Status
TC-05.1	Save and apply (REQ-05, AC-02.1)	1. As `rev_acme`, open S4, select V1 and V3, save. 2. Call E11. 3. Wait 60 s; run standard brief as `pm_acme`.	200 with `approved_at` within 2 s of server time; E11 lists V1, V3; the run calls only V1, V3.	Not Run
TC-05.2	Zero vendors (REQ-05, AC-02.2, failure)	1. Record E11. 2. In S4 clear all boxes and save.	Message "at least 1 vendor required"; 400 `at_least_one_vendor_required`; E11 unchanged.	Not Run
TC-05.3	Wrong role and bad input (REQ-05, edge)	1. PUT E4 as `pm_acme`. 2. As `rev_acme`, PUT 51 vendor IDs. 3. PUT one unknown UUID.	403 `reviewer_access_required`; 400; 404 `unknown_vendor` listing the ID.	Not Run

TC-06: Rate-limit handling (REQ-06, Must; owner Aditya Pawar)

ID	Description	Steps	Expected Result	Status
TC-06.1	Pause after two 429s (REQ-06, AC-01.3)	1. Set mock V2 to 429 always. 2. Run standard brief. 3. Compare server log and mock log timestamps.	V2 `delayed` ≤ 5 s after the 2nd 429; 0 calls to V2 for 60 s; V1, V3, V4 ranked.	Not Run
TC-06.2	Longer Retry-After (REQ-06, edge)	Set V2 to 429 with `Retry-After: 120`; run standard brief.	0 calls to V2 for 120 s.	Not Run
TC-06.3	Non-consecutive 429s (REQ-06, edge)	Set V2 to alternate 429, 200, 429, 200.	V2 never `delayed`; its models scored.	Not Run
TC-06.4	Real vendor on staging (REQ-06)	On staging, lower V1's limit with a test key at 2 requests/min; run standard brief.	Same behavior as TC-06.1 with the real API.	Not Run

TC-07 to TC-13: Should requirements

ID	Description	Steps	Expected Result	Status
TC-07.1	Vendor list (REQ-07; Allison)	As `rev_acme`, GET E7; open S4.	4 vendors, each with a non-empty `https://` policy link; the link opens in a new tab.	Not Run
TC-07.2	Missing policy link (REQ-07, failure)	Insert a vendor row with a null `retention_policy_url`.	Insert rejected by the NOT NULL constraint.	Not Run
TC-08.1	Select a model (REQ-08; Aditya)	After TC-01.1, POST E9 with rank-1 `entry_id`; GET E2.	200 with `selected_at`; E2 shows `selected_entry_id`.	Not Run
TC-08.2	Not selectable (REQ-08, failure)	POST E9 with a `delayed` entry; then on a `pending_review` report.	400 `entry_not_selectable`; 409 `report_not_released`.	Not Run
TC-09.1	Review hold (REQ-09; Aditya)	Run standard brief as `pm_gamma` (requires review); check E5 as PM and E12 as `se_1`.	Status `pending_review`; PM sees no report; `se_1` has `report_needs_review`.	Not Run
TC-09.2	Review and release (REQ-09)	As `se_1`, POST E8 moving 2 entries with notes, `release: true`; GET E5 as PM.	New order shown; `original_rank` kept; notes shown.	Not Run
TC-09.3	Review limits (REQ-09, failure)	POST E8 moving 3 entries; then a 501-character note; then `se_2` releases after `se_1`.	400 `too_many_reorders`; 400; 409 `already_released`.	Not Run
TC-10.1	Low-headroom warning (REQ-10; Allison)	Set mock V1 remaining-requests header to 9%; run standard brief.	One ops-webhook post ≤ 10 s with `type`, `vendor`, `remaining_pct`, `run_id`.	Not Run
TC-10.2	First 429 warning (REQ-10)	Set V2 to return one 429.	One `rate_limit_warning` for V2.	Not Run
TC-10.3	Deduplication (REQ-10, edge)	Trigger 20 warnings for V1 within 10 min.	Exactly 1 webhook post.	Not Run
TC-11.1	Every call logged (REQ-11; Aditya)	Run standard brief; compare mock log with `vendor_calls`.	100 rows; each written ≤ 2 s after the call; cost = tokens × price within $0.000001.	Not Run
TC-11.2	Failed call logged (REQ-11, edge)	Set V3 to `timeout` (25 s).	V3 rows have outcome `timeout`.	Not Run
TC-11.3	Log write fails (REQ-11, failure)	Block writes to `vendor_calls` during a run.	Run still completes; error logged; gap flagged for the operator.	Not Run
TC-12.1	History paging (REQ-12; Allison)	Seed 25 Acme reports; GET E3, then E3 with `next_cursor`.	20 newest first; then the remaining 5; `next_cursor` null.	Not Run
TC-12.2	Bad paging input (REQ-12, edge)	GET E3 with `limit=0`, `limit=101`, a tampered cursor.	400 `invalid_limit` twice; 400 `invalid_cursor`.	Not Run

ID	Description	Steps	Expected Result	Status
TC-12.3	Other company's report (REQ-12, failure)	As `pm_beta`, GET E2 with an Acme `report_id`, then with a random UUID.	Both 404 `report_not_found` with identical bodies.	Not Run
TC-13.1	Cancel (REQ-13; Aditya)	Start standard brief; after 10 s POST E6.	`cancelled`; 0 mock calls start > 5 s after cancel; no `reports` row.	Not Run
TC-13.2	Idempotent resubmit (REQ-13)	POST E1 twice with the same Idempotency-Key 1 min apart.	Same `run_id`; mock call count unchanged by the second.	Not Run
TC-13.3	Cancel a finished run (REQ-13, edge)	POST E6 on a `released` run.	409 `run_not_active`.	Not Run

TC-14 to TC-16: Could requirements (run only if built by M5)

ID	Description	Steps	Expected Result	Status
TC-14.1	Expiry reminder (REQ-14)	Save list expiring in 8 days; advance clock 1 day; run reminder job twice.	One email to `rev_acme`; `reminder_sent_at` set; a 91-day expiry is rejected with 400.	Not Run
TC-15.1	List-active notice (REQ-15)	Save a list as `rev_acme`.	Each Acme PM gets `vendor_list_active` ≤ 60 s.	Not Run
TC-15.2	No email address (REQ-15, edge)	Remove the email of one Acme PM; save a list.	That PM gets the in-app notice only; a warning is logged.	Not Run
TC-16.1	Auto-release (REQ-16)	Leave a Gamma report in review; advance clock 24 h.	Report released; `review_overdue` to engineers.	Not Run
TC-16.2	Just before the limit (REQ-16, edge)	Advance clock 23 h 59 min; run the timeout job.	Report still `pending_review`; no notification.	Not Run

TC-17 to TC-22: Performance, cost and security

ID	Description	Steps	Expected Result	Status
TC-17.1	Run time, mock (NFR-05; Aditya)	Mock latency median 800 ms, p95 2.5 s; 20 consecutive standard runs at realistic size.	≥ 19 of 20 terminal ≤ 90 s.	Not Run
TC-17.2	Run time, real vendors (NFR-05)	Same 20 runs on staging.	≥ 19 of 20 terminal ≤ 90 s.	Not Run
TC-17.3	Slow vendor (NFR-05, edge)	Set V4 to 15 s per call.	Run terminal ≤ 90 s with warning `deadline_reached`.	Not Run
TC-18.1	Read latency (NFR-06; Allison)	Seed 10,000 reports; k6 with 20 VUs for 5 min over E2, E3, E5, E7, E11, E12.	p95 ≤ 500 ms per endpoint.	Not Run
TC-18.2	Deep paging (NFR-06, edge)	Page E3 to the 50th page.	Each page ≤ 500 ms.	Not Run
TC-19.1	Calls per run (NFR-07; Allison)	Run standard brief; count `vendor_calls`.	≤ 100 plus one per failed call.	Not Run
TC-19.2	Active-run cap (NFR-07, failure)	Start 4 runs for Acme within 5 s.	4th returns 409 `run_in_progress`.	Not Run
TC-19.3	Hourly cap (NFR-07, edge)	Start 21 runs in one hour (clock controlled).	21st returns 429 with `Retry-After`.	Not Run
TC-20.1	Role matrix (NFR-08; Aditya)	Call E1–E12 with each of 4 roles, no token, expired token, bad signature, wrong audience (96 calls).	100% match `auth_matrix.csv`.	Not Run
TC-20.2	Clock skew (NFR-08, edge)	Token expired 29 s ago; then 31 s ago.	Accepted; then 401.	Not Run
TC-20.3	Identity provider down (NFR-08, failure)	Block JWKS URL for 11 min.	Requests succeed for 10 min on cached keys, then 503 `auth_unavailable`.	Not Run
TC-21.1	PII corpus (NFR-09; Aditya)	Embed the 200 items of `pii_corpus.txt` (100 emails, 100 phone formats) in 20 briefs; run them; regex-scan mock log.	0 seeded values in any outbound prompt.	Not Run
TC-21.2	Log redaction (NFR-09)	After TC-21.1, grep application logs for prompt text, completions and `Bearer`.	0 matches.	Not Run
TC-21.3	Grader path (NFR-09, edge)	Inspect grader calls from TC-21.1.	Grader prompts also contain 0 seeded values.	Not Run
TC-22.1	TLS (NFR-10; Allison)	Run testssl.sh against staging API and Client.	Only TLS 1.2 and 1.3 offered.	Not Run
TC-22.2	Secrets in repo (NFR-10, failure)	Push a branch containing a fake API key.	CI gitleaks step fails; main stays at 0 findings.	Not Run
TC-22.3	Token storage (NFR-10)	Sign in via Playwright; read localStorage and sessionStorage.	No token present.	Not Run

TC-23 to TC-29: Usability, reliability and constraints

ID	Description	Steps	Expected Result	Status
TC-23.1	PM task, staging (NFR-11; Allison)	5 first-time users get the script “Find the best model for summarizing tickets under $3 per 1,000 requests and 2 s”; observer times from S1 load.	≥ 4 of 5 submit a valid brief and name the top model and one reason ≤ 3 min, unaided.	Not Run
TC-23.2	PM task, pilot (NFR-11, validation)	Repeat TC-23.1 with 5 Concentrate AI product managers.	Same threshold.	Not Run
TC-24.1	Reviewer task (NFR-12, Could)	5 first-time users approve 2 vendors from S4.	≥ 4 of 5 ≤ 2 min, unaided.	Not Run
TC-25.1	Error messages (NFR-13; Allison)	Trigger each of the 17 4xx codes from E1–E12 through the Client (mocked API responses).	Each shows its mapped plain-language message; 0 raw codes.	Not Run
TC-25.2	Network loss (NFR-13, edge)	Go offline in Playwright during polling for 130 s.	Banner with Retry; Retry resumes the same `run_id`.	Not Run
TC-26.1	Vendor 429 (NFR-14; Aditya)	20 runs with V2 in 429 mode.	V1, V3, V4 scored in 20 of 20.	Not Run
TC-26.2	Vendor timeout (NFR-14)	20 runs with V2 in `timeout` mode.	Others scored in 20 of 20.	Not Run
TC-26.3	Vendor 5xx (NFR-14)	20 runs with V2 returning 503.	Others scored in 20 of 20.	Not Run
TC-26.4	Vendor 401 (NFR-14)	20 runs with V2 returning 401.	Others scored in 20 of 20; one operator alert.	Not Run
TC-26.5	All vendors fail (NFR-14, failure)	Set V1–V4 to 503.	Run `failed` with `all_vendors_unavailable`; no report.	Not Run
TC-27.1	Report write fails (NFR-15; Allison)	Block database writes during `storeReport`.	One retry; run `released_unsaved`; E5 shows `save_status: not_saved` and the shortlist.	Not Run
TC-27.2	Restart mid-run (NFR-15, failure)	Kill the server 20 s into a run; restart.	Run `failed` with `run_interrupted` ≤ 30 s after startup.	Not Run
TC-27.3	No stuck runs (NFR-15)	After TC-27.2, count runs in `queued` or `dispatching`.	0.	Not Run
TC-28.1	Notification retry (NFR-16, Could)	Point email at a failing stub; queue one notification.	5 retries at 30/60/120/240/480 s, then `failed`; a duplicate key adds no row.	Not Run
TC-29.1	Reasoning wording (CON-04; Allison, Todd Liebermann)	Generate 50 reasoning strings from varied briefs; scan for the banned list (“better than”, “beats”, “best model”, vendor-vs-vendor wording); Todd reviews.	0 banned phrases; Todd signs off.	Not Run

Constraint coverage. CON-01: TC-11, TC-13, TC-19. CON-02: TC-04, TC-20, TC-21. CON-03: TC-02, TC-01.5. CON-04: TC-12.3, TC-29. CON-05 has no test case; it is checked at milestone reviews.

5. Test Environment

Item	Development	Staging	Pilot
Purpose	Unit, integration, system tests on every push	Real-vendor, load and usability tests before reviews	Validation with Concentrate AI users (M4–M6)
Hosts	macOS 15 laptops (Apple silicon) and GitHub Actions `ubuntu-24.04`	Managed container platform (Concentrate AI account), 1 instance, 1 vCPU / 2 GB	Same platform, separate project
Runtime	Node.js 22 LTS (pinned in `.nvmrc`), Docker 27	Same Docker image as CI	Same image, tagged release
Database	PostgreSQL 16 (`postgres:16` Docker image), migrated with node-pg-migrate	Managed PostgreSQL 16, TLS required	Managed PostgreSQL 16, separate database
Vendors	Mock vendor server with modes `ok`, `429`, `timeout`, `5xx`, `401`, `empty`, `truncated`, headers configurable	Real OpenAI and Anthropic test keys with spend caps	Real keys under Concentrate AI’s account
Identity	Local OIDC stub issuing signed test tokens	Identity-provider staging tenant (OQ-04)	Production tenant, pilot users only
Browsers	Chromium, Firefox and WebKit bundled with the Playwright version pinned in `package-lock.json`	Chrome, Safari and Firefox current stable; versions recorded per run	Users’ own browsers; version recorded in session sheet
Network	Local; Toxiproxy for faults; Playwright offline and Slow 3G profiles	Public internet from Cincinnati	Users’ own networks
Tools	vitest, Playwright, k6, Toxiproxy, gitleaks	k6, testssl.sh	Observer sheet, stopwatch

Seeded data (`npm run seed:test`, deterministic): client companies Acme (V1–V4 approved), Beta (no approvals), Gamma (requires review); one user per role per company plus `se_1`, `se_2`, `op_1`; vendors V1–V4 with policy links; 10 models with fixed prices; `grades_a.json`, `expected_order_a.json`, `a2_reference.csv`, `auth_matrix.csv`, `pii_corpus.txt` (200 items); 20 briefs; and, for load tests only, 10,000 reports.

6. Pass and Fail Criteria

Each test case. Pass when every condition in Expected Result holds exactly, including each threshold. Fail when any condition does not hold or the system crashes. Blocked when the environment, not the system, prevents the run (for

example a vendor outage on staging); a blocked case is rerun and never counted as a pass. Not Run means it has not been executed. A failure is logged as a GitHub issue with severity:

- **Sev 1:** data leaves to an unapproved vendor, PII or tokens leak, another company's data is visible, or stored data is lost or corrupted.
- **Sev 2:** a Must requirement fails, or a metric in Section 3 misses its threshold.
- **Sev 3:** a Should requirement fails, with a workaround.
- **Sev 4:** cosmetic.

Metrics. A metric passes when the measured value meets the Section 3 threshold on the stated number of trials. For sampled metrics (NFR-05, NFR-14, NFR-11) the sample is fixed in advance and not extended to make a result pass.

The plan as a whole passes at M6 when:

1. 100% of Must cases (TC-01 to TC-06, TC-17, TC-20, TC-21, TC-26) pass.
2. At least 90% of Should cases pass, and every failing one has a logged Sev 3 issue.
3. Every Section 3 metric for Must and Should requirements meets its threshold.
4. 0 open Sev 1 or Sev 2 defects.
5. Validation cases TC-01.5, TC-23.2 and TC-29.1 pass, or the advisors accept a documented exception.

It fails if any of these is not met. Could cases do not count either way.

7. Schedule

Unit and integration tests run on every push during sprints; system and acceptance tests run before each review and demo. Order follows risk (Section 1).

Milestone (date)	Test activity	Owner	Evidence
MI-1 Dev environment (2026-12-04)	Mock vendor server, OIDC stub, seed script and CI skeleton built; smoke call to each real vendor	Aditya Pawar	Mock server in repo; CI run log; vendor smoke output
M0 CI and config (2027-01-20)	Unit tests for config and logging; gitleaks in CI (TC-22.2)	Allison Oh	Green CI run; failing run on planted key
M1 Walking skeleton (2027-01-29)	Integration: auth matrix (TC-20), store schema and constraints (TC-07.2), brief validation (TC-03)	Aditya Pawar, Allison Oh	CI report with case IDs
M2 Real vendor benchmarks (2027-02-16)	Integration: approved vendors only (TC-04), rate limits (TC-06.1–06.3), usage log (TC-11), isolation (TC-26); staging TC-06.4	Aditya Pawar	CI report; mock and staging logs
M3 Ranked shortlist (2027-03-09)	Unit and integration: ranking and reasoning (TC-01.1–01.4, TC-02); load TC-17.1, TC-17.3; cost caps TC-19	Aditya Pawar	A2 test report; k6 and run-time CSV
M4 MVP advisor demo (2027-03-25)	System: UC-01 end to end in Playwright; usability TC-23.1; ranking validation TC-01.5; wording TC-29	Allison Oh	Playwright report; session sheets; advisor notes
M5 Feature complete (2027-04-10)	System: approval (TC-05), review and selection (TC-08, TC-09), notifications (TC-10), history (TC-12), cancel (TC-13), errors (TC-25); Could cases if built	Allison Oh	Playwright report; issue list
M6 Validation complete (2027-04-28)	Full regression; staging TC-17.2, TC-18, TC-21, TC-22, TC-27; stability TC-01.6; pilot TC-23.2	Allison Oh (tests), Aditya Pawar (load)	Test report with pass/fail per case and metric; advisors sign off
M7 Final presentation and handoff (2027-04-30)	Report results; hand over test suite	Both	Final report section; repository

8. Definition of Done

A feature is done when:

- ☐ Its test cases in Section 4 are written as automated tests (or a session script for usability) and pass in CI or on staging.
- ☐ Every edge and failure case in its group passes.
- ☐ The code has been reviewed and approved by the other team member.
- ☐ Unit-test line coverage for its module is at least 80%.
- ☐ The API, configuration or screen change is reflected in the Technical Specification and README.
- ☐ It has 0 open Sev 1 or Sev 2 defects.
- ☐ Its traceability-matrix row is updated (status and test-case IDs).

The project is done when:

- ☐ The plan-level pass criteria in Section 6 are met.
- ☐ Every Must and Should requirement has a passing test or an advisor-accepted exception.

- [ ] Every Section 3 metric has a recorded measured value.
- [ ] The test report, Requirements Document, Technical Specification and this plan are at matching versions.
- [ ] 0 open Sev 1 or Sev 2 defects; open Sev 3 and 4 issues are listed in the handoff.
- [ ] Jillian Aurisano and Todd Liebermann have signed off on the M6 test report.